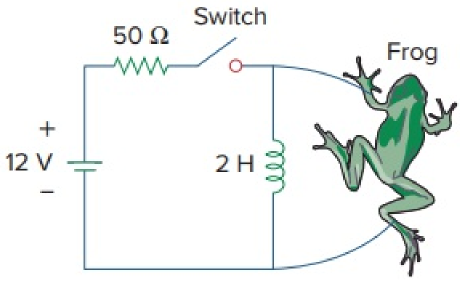
Chapter 7, Problem 91CP

Problem

The circuit in Fig. 7.150 is used by a biology student to study “frog kick.” She noticed that the frog kicked a little when the switch was closed but kicked violently for 5 s when the switch was opened. Model the frog as a resistor and calculate its resistance. Assume that it takes 10 mA for the frog to kick violently.

Fig. 7.150:



Step-by-step solution

Step 1 of 1

$$i_0(0) = \frac{12}{50} = 240\ mA \qquad i(\infty) = 0$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]\, e^{\frac{-t}{\tau}}$$

$$= 240\ e^{\frac{-t}{\tau}}$$

$$\tau = \frac{L}{R} = \frac{2}{R}$$

$$i(t_0) = 10 = 240 e^{\frac{-t_0}{\tau}}$$

$$e^{\frac{t_0}{\tau}} = 24 \rightarrow t_0 = \tau\, ln(24)$$

$$\tau = \frac{t_0}{ln(24)} = \frac{5}{ln(24)} = 1.573 = \frac{2}{R}$$

$$R = \frac{2}{1.573} = 1.271\ \Omega$$